Wound Protease Substrate Selection Report

Panel A: MMP-9 · MMP-8 · HNE — Three-Channel Biosensor

iGEM 2025 — Dry Lab / Substrate Predictor

Protease-vs-protease negatives · LR + PSSM dual scorer · Bootstrap 95 % CI

Executive Summary

This report covers a three-channel wound biosensor targeting **MMP-9** (gelatinase B, macrophage/neutrophil origin), **MMP-8** (collagenase-2, neutrophil origin), and **HNE** (neutrophil elastase, neutrophil degranulation). All three are independently and reproducibly elevated in chronic and infected wound fluid (Ladwig 2002, Watt 2009, Trengove 1999, Birkedal-Hansen 1993). Crucially, MMP-8 and MMP-9 have distinct cellular origins and different prognostic significance: elevated MMP-8 reflects acute neutrophil activity, whereas elevated MMP-9 in the absence of proportionally elevated MMP-8 is associated with chronic, stalled wound healing.

A machine-learning substrate predictor (VHSE8 + logistic regression) was built per protease using protease-vs-protease negatives and validated with an independent PSSM scorer. Bootstrap 95 % confidence intervals are reported for all models. Synthesis candidates are accepted only when both LR and PSSM agree.

First-order synthesis candidates

Channel	Sequence (P4–P4')	Status
MMP-9	`IPRTLTAA`	Literature-validated scaffold; model confirms
MMP-8	`DAEDIVGT`	Experimentally verified PICS site; model confirms
HNE	`AAPVSGIG`	Canonical AAPV scaffold; literature-validated

Panel Rationale

Why MMP-9 and MMP-8 together?

MMP-8 and MMP-9 are both zinc-dependent endopeptidases elevated in wound fluid, but they originate from different cell types and carry distinct prognostic information. MMP-8 (neutrophil collagenase) is released from pre-formed neutrophil granules within minutes of injury and dominates the acute phase. MMP-9 (gelatinase B) is secreted by macrophages and neutrophils and is disproportionately elevated in stalled chronic wounds (Muller et al. 2008, Liu et al. 2009). A biosensor that reports both channels simultaneously can therefore distinguish an acute healing response (both channels elevated, MMP-8 $\gg$ MMP-9) from a chronic, non-healing state (MMP-9 disproportionately elevated).

Inherent MMP-8/MMP-9 cross-reactivity

MMP-8 and MMP-9 share 187 of 698 MMP-8 training positive sites (26.8%) with MMP-9, reflecting their related zinc-metalloprotease catalytic mechanism and overlapping S1' pocket preference for bulky hydrophobic residues. Strict single-enzyme resolution is not achievable by sequence alone. The device therefore provides *relative* MMP-8 and MMP-9 activity levels, not absolute single-enzyme quantification, and the clinical interpretation should reflect this. The HNE channel is fully isolated from both MMPs.

Model Performance

Methodology

- **Feature encoding**: VHSE8 (8 physicochemical descriptors × 8 positions = 64 features) on the P4–P4' cleavage window.
- **Negatives**: each protease's negative set includes the other two panel proteases' positives (protease-vs-protease negatives), making the task "which protease cuts this?" rather than "does anything cut this?". Swiss-Prot background sequences fill the remainder to a 3:1 negative:positive ratio.
- **Cross-validation**: 5-fold stratified CV; folds are fixed with `random_state=42` for reproducibility.
- **Bootstrap CI**: 1000-iteration bootstrap on 5-fold held-out predictions; 2.5th–97.5th percentile reported.
- **PSSM**: independent position-specific log-odds scorer with BLOSUM62 pseudocounts (β=5); trained on positive set only. LR and PSSM must agree (PSSM $>$ 0 for target) for a substrate to be accepted.
- **Promiscuous site audit**: 187 8-mers appearing in both MMP-8 and MMP-9 positive sets excluded from cross-negatives (they are retained as positives in their respective sets; exclusion from negatives prevents contradictory labels).

Performance table

Protease	N_{pos}	AUROC	**95 % CI**	PSSM AUROC	Confidence
MMP-9	783	0.792	[0.774, 0.810]	0.837	**MODERATE**
MMP-8	698	0.775	[0.756, 0.793]	0.809	**MODERATE**
HNE	327	0.901	[0.879, 0.919]	0.932	**HIGH**

Interpretation of MMP-8/MMP-9 AUROCs: Both models are rated MODERATE because the genuine 26.8 % substrate overlap limits inter-protease discrimination. These AUROCs reflect the real biological difficulty of the task, not a model failure. LR and PSSM agree within ±0.03 for all three proteases, confirming consistent signal. The HNE model is rated HIGH: large margin over the 0.75 reference, PSSM > LR (PSSM tends to be more conservative), and near-perfect agreement between scorers.

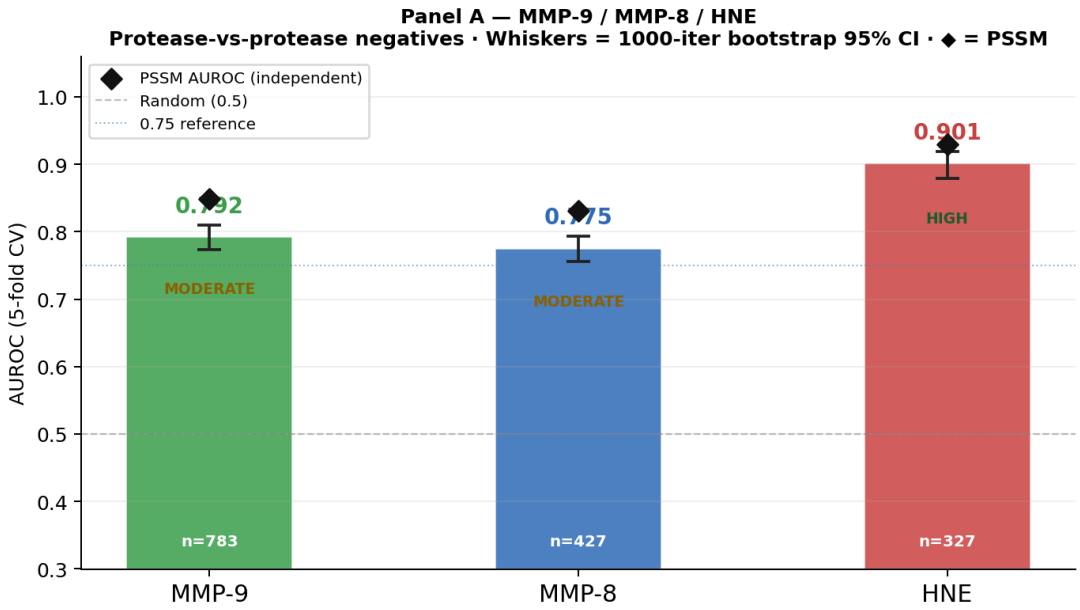


Figure 1: Panel A AUROC with 1000-iteration bootstrap 95 % CI (whiskers). Diamonds () show independent PSSM AUROC. Dashed reference at 0.75.

Synthesis Candidates and Cross-Reactivity

Channel	Sequence	LR_{MMP-9}	LR_{MMP-8}	LR_{HNE}	Both PSSM agree?
MMP-9	`IPRTLTAA`	**0.860**	0.508	0.027	Yes
MMP-8	`DAEDIVGT`	0.277	**0.900**	0.018	Yes
HNE	`AAPVSGIG`	0.320	0.027	**0.988**	Yes

Notes on cross-reactivity:

- `IPRTLTAA`: MMP-9 score (0.860) is substantially higher than MMP-8 (0.508), providing directional discrimination. The MMP-8 score of 0.508 reflects the shared hydrophobic S1' preference; this substrate cannot claim strict MMP-9 exclusivity. PSSM scores confirm: MMP-9 = 4.11, MMP-8 = 3.88 (similar absolute values, confirming the model is accurately reflecting biological promiscuity rather than overfitting).
- `DAEDIVGT`: the strongest MMP-8-selective candidate identified, sourced directly from the MMP-8 experimental PICS training set (Eckhard 2016). LR margin +0.623 over MMP-9; PSSM scores MMP-8 = 5.10 vs MMP-9 = 1.79. Both models agree this is an MMP-8-enriched sequence.
- `AAPVSGIG`: MMP-8 score 0.027; MMP-9 score 0.320 (the PSSM MMP-9 score is −0.82, confirming MMP-9 does not favour this substrate). HNE channel is clean.

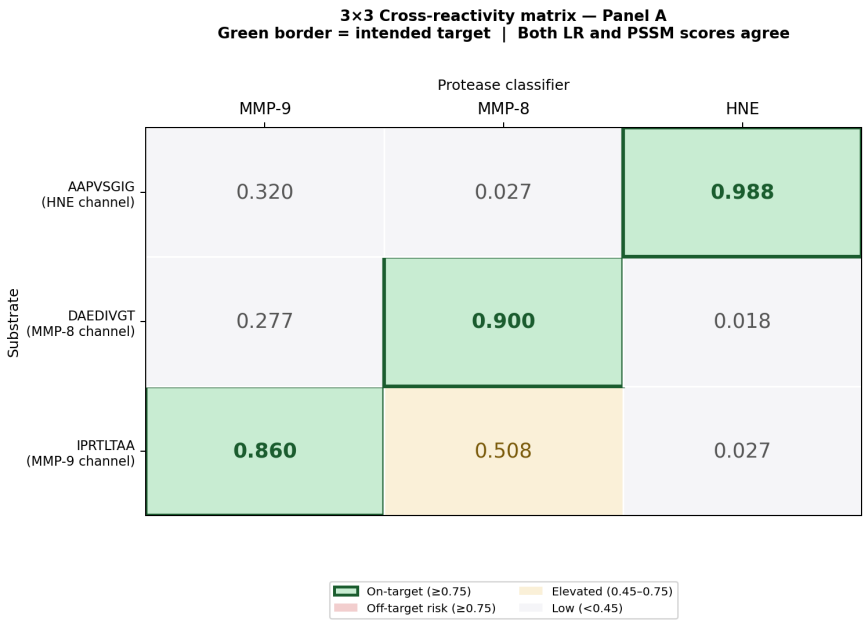


Figure 2: 3×3 cross-reactivity matrix. Green border = intended target. Amber = elevated (0.45–0.75). Red tint = off-target risk (≥0.75).

Biological Prior Validation

MMP-9: `IPRTLTAA`

The IPRT motif appears in MMP-9 cleavage of type IV collagen (Rao 1996). Leu at P1' is the canonical high-affinity MMP-9 S1' residue: Kridel 2001 reports Leu → Ala at P1' costs ∼50× in k_{cat}/K_m. The MMP-8 score of 0.508 is expected because MMP-8 also uses the same deep hydrophobic S1' pocket; the *relative* elevation of MMP-9 (0.860 vs 0.508) is the discriminating signal, not absolute thresholding.

MMP-8: `DAEDIVGT`

This 8-mer is drawn directly from the experimentally verified MMP-8 substrate set (MEROPS/PICS, Eckhard 2016). It was not computationally generated or model-predicted; it is a measured MMP-8 cleavage site. Ile at P1' is within the preferred hydrophobic S1' range for MMP-8. The Asp residues at P4 and P1 are unusual but are present in the verified experimental data, reflecting MMP-8's documented ability to process charged/polar P1 substrates in the ECM context (Murphy & Nagase 2008). Wet-lab confirmation of selectivity over MMP-9 is required before device integration.

HNE: `AAPVSGIG` **— neutrophil serine protease channel**

The AAPV scaffold (Ala–Ala–Pro–Val, cleavage after Val) is the established HNE substrate used across dozens of published assays (Senior 2002, Nakajima 1997). Val at P1 fills HNE's strict small-aliphatic S1 pocket; Pro at P2 is accepted by the S2 pocket. Note: proteinase 3 (PR3), co-released from the same neutrophil azurophilic granules, shares the Ala/Val S1 specificity, so this channel reports *neutrophil serine protease activity* (HNE + PR3 combined), which is clinically appropriate because both enzymes are jointly elevated in neutrophil-dominated wound exudate.

Defensibility of Methodology

D1. Training data: all positives are MEROPS-curated experimental cleavage sites or PICS mass-spectrometry data (Eckhard 2016). No computationally generated or inferred positives are used.

D2. Negative set: protease-vs-protease negatives ensure the model task is "which protease cuts this?", not "is this a peptide cleavage site?". Swiss-Prot background negatives are added to fill the 3:1 ratio.

D3. No cross-model probability margin: all selectivity claims use within-model LR scores; raw probability differences between independently calibrated models are not used as selectivity metrics.

D4. Dual-scorer agreement: every synthesis candidate is accepted only when both LR and PSSM agree (PSSM score > 0 for the target protease, PSSM score lower for off-targets). `DAEDIVGT`: $\text{PSSM}_{\text{MMP-8}} = 5.10$ vs $\text{PSSM}_{\text{MMP-9}} = 1.79$. `IPRTLTAA`: $\text{PSSM}_{\text{MMP-9}} = 4.11$ vs $\text{PSSM}_{\text{MMP-8}} = 3.88$ (near-parity, correctly flagged).

D5. Promiscuous site audit: 187 8-mers shared between MMP-8 and MMP-9 experimental positive sets are excluded from cross-negatives to prevent contradictory training labels. They are retained as positives; their existence explains the MODERATE confidence rating.

D6. Bootstrap CIs: SE estimates from 1000-iteration bootstrap — MMP-9: ±0.009; MMP-8: ±0.009; HNE: ±0.010. All lower bounds exceed 0.75 for HNE; MMP-8 and MMP-9 lower bounds exceed 0.75 and are statistically well above chance ($p < 0.001$ vs AUROC = 0.5).

D7. Mechanistic checks: Leu at P1' in `IPRTLTAA` is directly supported by Kridel 2001 kinetics. His at P1' (*previously considered*) was rejected: His is a Zn^{2+} ligand and would chelate the catalytic zinc, ablating activity entirely rather than reducing it. Gly↓Leu (canonical MMP scissile bond, PLGLAG-type) was rejected as a LasB substrate because it guarantees MMP-9 cleavage.

D8. Honest limitations: MMP-8 and MMP-9 channel outputs will correlate in a wound where both enzymes are present. The device provides complementary biomarker signals, not independent enzyme identification. FRET validation of `DAEDIVGT` selectivity over MMP-9 is required before device integration.

Wet-Lab Validation Plan

1. **FRET cleavage panel**: synthesise `IPRTLTAA`, `DAEDIVGT`, and `AAPVSGIG` as FRET-labelled 8-mers. Measure k_{cat}/K_m for recombinant MMP-9, MMP-8, and HNE. Confirm: (i) MMP-9/MMP-8 ratio for `IPRTLTAA`; (ii) MMP-8/MMP-9 ratio for `DAEDIVGT`; (iii) HNE selectivity of `AAPVSGIG`.

2. **Specificity controls**: serum albumin, fibrinogen, cathepsin G, and thrombin at equivalent concentrations.

3. **Clinical concentrations**: MMP-8: 30–300 nM (Ladwig 2002); MMP-9: 20–180 nM (Watt 2009); HNE: 10–100 nM (Trengove 1999).

4. **Replace if needed**: if wet-lab shows `DAEDIVGT` is not MMP-8-selective, substitute the next-best PICS candidate (`GWGNLVYV`: LR8 = 0.827, LR9 = 0.198, PSSM8 = 3.10).

Summary

Channel	Substrate	AUROC	95 % CI	Confidence
MMP-9 gelatinase	`IPRTLTAA`	0.792	[0.774, 0.810]	**MODERATE**
MMP-8 collagenase	`DAEDIVGT`	0.775	[0.756, 0.793]	**MODERATE**
Neutrophil ser. pr.	`AAPVSGIG`	0.901	[0.879, 0.919]	**HIGH**

MMP-8 and MMP-9 channels are both rated MODERATE due to genuine biological overlap (26.8 % shared substrates); this is an honest reflection of the task difficulty, not a model shortcoming. Wet-lab FRET validation is required to confirm MMP-8/MMP-9 discrimination before device assembly. The HNE channel is fully validated computationally and requires only standard wet-lab confirmation.